

★ DAILY pH + METAL PROPERTIES ★

GS | Ch04 Chem B + Bio A | Topic 1 of 12

► Meaning in one line

Stomach/soil/rain , lustre/malleable.

► Key points

Point	Example
pH life: stomach acid + soil + rain.	antacid + lime.
Metals: lustre + malleable + ductile + conduct.	gold + copper.

★ IMPORTANT FOR EXAM

Reactivity + series

⚠ TRAP - read twice

Lab-only + 'pH asked' + life-miss + trait-miss

✓ **Try yourself:** Life + trait + pH + metal + apply? **Answer:** Daily + props.

★ REACTIVITY + SERIES ★

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► Meaning in one line

Tendency , K Na Ca Mg Al Zn Fe.

► Key points

Point	Example
Reactivity: ease of reaction.	Na explodes in water.
Series high-yield: K Na Ca Mg Al Zn Fe Pb H Cu + displaces below.	Zn + CuSO ₄ .

★ IMPORTANT FOR EXAM

Corrosion + prevention

⚠ TRAP - read twice

Metal-only + 'reacts asked' + order-miss + displace-miss

✓ **Try yourself:** Order + displace + series + exam? **Answer:** Series + displace.

★ CORROSION + PREVENTION ★

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► Meaning in one line

Rust needs air , water , coat/galvanize.

► Key points

Point	Example
Corrosion: iron + air + water = rust.	gates + ships.
Prevent: oil/paint/galvanize/alloy.	zinc coat + stainless.

★ IMPORTANT FOR EXAM

Alloys + uses

⚠ TRAP - read twice

Rust-only + 'corrode asked' + needs-miss + shield-miss

✓ **Try yourself:** Needs + shield + rust + coat + block? **Answer:** Rust + prevent.

★ ALLOYS + METAL/NONMETAL USES ★

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► Meaning in one line

Mix wins , brass/steel , wires/foil.

► Key points

Point	Example
Alloys: brass + bronze + steel.	bells + tools.
Uses: Cu wires + Al foil + S tyres + graphite.	daily items.

★ IMPORTANT FOR EXAM

Soap + bleach

⚠ TRAP - read twice

Pure-only + 'alloy asked' + mix-miss + use-miss

✓ **Try yourself:** Mix + use + alloy + daily? **Answer:** Alloy + uses.

★ SOAP + BLEACHING AGENTS ★

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► Meaning in one line

Fat , alkali vs synthetic , whiten.

► Key points

Point	Example
Soap: fat + alkali + fails in hard water. Detergent works.	-
Bleach: chlorine + whiten + disinfect.	clothes + water.

★ IMPORTANT FOR EXAM

Fertilizers + fuels

⚠ TRAP - read twice

Soap-only + 'clean asked' + hard-miss + agent-miss

✓ **Try yourself:** Hard + agent + clean + whiten + sort? **Answer:** Soap + bleach.

★ FERTILIZERS + FUELS ★

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► Meaning in one line

NPK , burn value.

► Key points

Point	Example
Fertilizers: NPK + urea + potash.	crops need N.
Fuels: petrol/diesel/CNG/LPG + calorific.	kitchen + vehicles.

★ IMPORTANT FOR EXAM

Polymers + household

⚠ TRAP - read twice

Name-only + 'fuel asked' + NPK-miss + burn-miss

✓ **Try yourself:** Feed + burn + NPK + fuel + sort? **Answer:** Fertilizer + fuel.

★ POLYMERS + HOUSEHOLD CHEMICALS ★

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► Meaning in one line

Repeat units , plastic/rubber , daily.

► Key points

Point	Example
Polymers: repeat units + plastic/rubber/fibre.	bags + tyres.
Household: cleaners/phenyl/naphthalene.	kitchen + toilet.

★ IMPORTANT FOR EXAM

Hard water + fire triangle

⚠ TRAP - read twice

Plastic-only + 'polymer asked' + repeat-miss + daily-miss

✓ **Try yourself:** Repeat + daily + plastic + home + sort? **Answer:** Polymer + home.

★ HARD WATER + FIRE TRIANGLE ★

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► Meaning in one line

Ca/Mg scale , fuel , oxygen , heat.

► Key points

Point	Example
Hard: Ca/Mg + soap scum + scale + boil/ion-exchange.	-
Fire: fuel + oxygen + heat + remove one.	blanket + CO ₂ .

★ IMPORTANT FOR EXAM

Cell theory + prokaryote

⚠ TRAP - read twice

Water-only + 'hard asked' + ion-miss + leg-miss

✓ **Try yourself:** Ions + legs + scale + fire? **Answer:** Hard + triangle.

★ CELL THEORY + PROKARYOTE ★

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► Meaning in one line

Unit of life , no-nucleus.

► Key points

Point	Example
Theory: cell = unit + all from cells.	plants + animals.
Prokaryote: no nucleus + no organelles + bacteria.	E. coli.

★ IMPORTANT FOR EXAM

Plant/animal + membrane

⚠ TRAP - read twice

Cell-only + 'cell asked' + unit-miss + nucleus-miss

✓ **Try yourself:** Unit + nucleus + simple + cell + sort? **Answer:** Theory + prokaryote.

★ PLANT/ANIMAL CELL + MEMBRANE ★

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► Meaning in one line

Wall vs no-wall , gate.

► Key points

Point	Example
Plant: wall + big vacuole + chloroplast. Animal: no wall + small vacuole.	-
Membrane selective + wall rigid support.	osmosis gate.

★ IMPORTANT FOR EXAM

Nucleus + mitochondria

⚠ TRAP - read twice

Cell-only + 'plant asked' + wall-miss + gate-miss

✓ **Try yourself:** Wall + gate + plant + guard + sort? **Answer:** Cell + membrane.

★ NUCLEUS + MITOCHONDRIA ★

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► Meaning in one line

Control , power.

► Key points

Point	Example
Nucleus: DNA + control.	brain of cell.
Mitochondria: ATP + powerhouse + aerobic.	muscle rich.

★ IMPORTANT FOR EXAM

Ribosome + ER/Golgi

⚠ TRAP - read twice

Part-only + 'organelle asked' + DNA-miss + ATP-miss

✓ **Try yourself:** DNA + ATP + control + power + link? **Answer:** Nucleus + mito.

★ RIBOSOME + ER/GOLGI ★

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► Meaning in one line

Protein , transport , pack.

► Key points

Point	Example
Ribosome: protein factory.	free + bound.
ER transports + Golgi packs/ships.	post office.

★ IMPORTANT FOR EXAM

Till end: division + systems!

⚠ TRAP - read twice

Organelle-only + 'cell asked' + make-miss + pack-miss

✓ **Try yourself:** Make + pack + road + factory + link? **Answer:** Ribosome + ER/Golgi.